

THE LANCET

Supplementary appendix

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

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SUPPLEMENTARY APPENDIX

PEACE-1 STUDY

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2. Supplementary methods

2.1 Ethics and oversight

The Independent Ethics Committee (CPP Ile de France VII, Hôpital de Bicêtre, 78, rue du Général Leclerc, 94270 Le Kremlin Bicêtre – France; Email: cyp.idf.7-bicetre@orange.fr; Phone: +33 (0)1 45 21 28 46; Fax: +33 (0)1 42 34 86 11) reviewed and approved the study documents, including the initial protocol, all subsequent amendments as well as all information and documents provided to each and every patient included in this study. The initial protocol received a positive review on July 10TH, 2013.

An Independent Data Monitoring Committee (IDMC) and a Steering Committee were set up to ensure trial oversight, enforce ethic, protect all patients, and evaluate benefit/risk ratios based on the independent review of all scientific results during and after study completion. Steering Committee members are presented in section 3 below.

Each and every eligible patient was informed about the risk and eventual benefit of the study they were about to join. They were given enough time to understand the procedure, ask questions and consider their options before being required to sign a dedicated informed consent form ahead of being centrally randomized.

2.2 Missing data

For rPFS and OS, missing values on the outcome were handled by means of censoring. Notably, recent vital status data (obtained within 3 months of the cut-off date) were available for 97% of the study patients. All the efficacy analyses were done on the 1172 patients from the ITT population, except the Cox model analysis used for the rPFS analysis (there were 6 patients with missing data for the stratification factor “Disease burden”, hence discarded from the analysis). These 6 missing data were recovered at time of OS analysis so that the Cox model applied to this analysis took into account the whole study ITT population (n=1172).

3. Steering committee members

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CRA stands for Clinical research associate; EORTC, European Organization for Research and Treatment of Cancer.

4. Inclusion and non-inclusion criteria

4.1 Inclusion criteria

1. Histologically or cytologically confirmed adenocarcinoma of the prostate,
2. Metastatic disease documented by a positive bone scan (any technique) or CT scan or an MRI. For patients with nodal metastases only, only patients with extra-pelvic enlarged lymph nodes (lymph nodes located above the iliac bifurcation) can be included if they have either:
 - At least one extra-pelvic lymph node ≥ 2 cmor
 - Extra-pelvic lymph node (s) ≥ 1 cm if the patient also have at least one pelvic lymph node ≥ 2 cm
3. Patients with ECOG ≤ 1 (patient with PS 2 due to bone pain can be accrued in the trial),
4. Life expectancy of at least 6 months,
5. Male aged ≥ 18 years old and ≤ 80 years old,
6. Hematology values:
 - Hemoglobin ≥ 10.0 g/dL,
 - Platelet count $\geq 100,000/\mu\text{L}$,
 - Neutrophil ≥ 1500 cells/mm³
7. Biochemistry values:
 - Renal function: serum creatinine $< 1.5 \times \text{ULN}$ or a calculated creatinine clearance ≥ 60 mL/min,
 - Serum potassium ≥ 4 mmol/L,
 - Liver function:
 - Serum bilirubin $\leq 1.5 \times \text{ULN}$ (except for patients with documented Gilbert's disease);
 - AST and ALT $\leq 1.5 \times \text{ULN}$ (and $\leq 5 \times \text{ULN}$ in case of liver metastases);
 - ALP $\leq 2.5 \times \text{ULN}$ (in case of bone metastasis, ALK-P $< 1000\text{U/L}$ if bilirubin is normal)
8. Patients must have received ADT for a maximum of 3 months before randomization and there must be a minimum of 6 weeks between the start of ADT and the start of Docetaxel.
9. Patients willing and clinically fit to receive Docetaxel, which is defined by the following:
 - Patients respecting all inclusion and exclusion criteriaAnd
 - Patients with no contraindication to docetaxel according to the SmPC of the drugAnd
 - Patients presenting all medical requirements to receive docetaxel according to the investigator's opinion.
10. Patients might have received previous radiation therapy directed to bone lesions,
11. Patients able to take oral medication,
12. Patients who have received the information sheet and signed the informed consent form,
13. Male patients who will receive Docetaxel and/or Abiraterone acetate and have partners of childbearing potential and/or pregnant partners must use a method of birth control in addition to an adequate barrier protection (condoms) as determined to be acceptable by the study doctor during the treatment period and for 4 weeks after the last dose of abiraterone acetate and/or for 6 months after the last dose of Docetaxel,
14. Patients must be willing and able to comply with scheduled visits, treatment plan, laboratory tests and other study procedures,
15. Patients with a public or a private health insurance coverage, according to local laws for participation in clinical trials.

4.2 Non-inclusion criteria

1. Patients with previous definitive local treatment directed to the prostate primary cancer (radiotherapy, brachytherapy, radical prostatectomy, ultrasound, cryotherapy, or other). A previous trans-urethral resection of the prostate (TURP) and previous local treatments of metastases are allowed,
2. Prior cytotoxic chemotherapy or biological therapy for the treatment of prostate cancer,
3. Any chronic medical condition requiring a higher dose of corticosteroid than 5 mg prednisone/prednisolone twice daily,
4. Active infection or other medical condition for which prednisone/prednisolone (corticosteroid) use would be contraindicated,
5. Previously treated with ketoconazole for prostate cancer for more than 7 days,
6. Prior systemic treatment with an azole drug (e.g., fluconazole, itraconazole) within 4 weeks of randomization,
7. Hypertension not controlled by an anti-hypertensive (systolic BP ≥ 160 mmHg or diastolic BP ≥ 95 mmHg; 3 consecutive measures taken 5 minutes apart),
8. Severe or moderate hepatic impairment (Child – Pugh class C or B)
9. Active or symptomatic viral hepatitis or chronic liver disease (except Gilbert's disease),
10. History of pituitary or adrenal dysfunction,
11. Clinically known significant heart disease as evidenced by myocardial infarction, or arterial thrombotic events in the past 6 months, severe or unstable angina, or New York Heart association (NYHA) Class II-IV heart disease or cardiac ejection fraction measurement of $< 50\%$ at baseline,
12. Atrial Fibrillation, or other cardiac arrhythmia requiring therapy,
13. Patient with unstable pulmonary disease (eg., Pulmonary embolism)
14. Pathological finding consistent with small cell carcinoma of the prostate,
15. History of malignancy, except non-melanoma skin cancer, with a $\geq 30\%$ probability of recurrence within 24 months,
16. Known allergies, hypersensitivity or intolerance to the study drugs or excipients or Docetaxel,
17. Administration of an investigational therapeutic within 30 days of randomization,
18. Patients already included in another therapeutic trial involving an experimental drug (patient in a non-experimental trial with no modification of the patient's care can be included),
19. Patients with significantly altered mental status prohibiting the understanding of the study or with psychological, familial, sociological or geographical condition potentially hampering compliance with the study protocol and follow-up schedule or any condition which, in the opinion of the investigator, would preclude participation in this trial. Those conditions should be discussed with the patient before registration in the trial,
20. Individual deprived of liberty or placed under the authority of a tutor.
21. Patients with impaired vision should undergo a prompt and complete ophthalmologic examination. Patients with Cystoid Macular Oedema cannot be included due to a potential risk of deterioration associated with docetaxel., the patient should not receive docetaxel.
22. Concomitant use of strong CYP3A4 inhibitors (clarithromycin, indinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin)

5. Amendments to the study design and statistical analysis plan

Since PEACE 1 initial protocol authorization on July 13TH, 2013, the treatment of men with *de novo* metastatic prostate cancer has drastically evolved. Consequently, the study design and statistical analysis plan had to be adapted accordingly. Changes to the study protocol and statistical analysis plan were made without any reference to the outcomes data collected during the study. All amendments to the protocol were validated by the Steering committee and the Independent Data Monitoring Committee before being submitted to dedicated regulatory authorities (depending on the country in which is located a specific participating site) ahead of implementation. Study amendments to the protocol that affected the course of the study and/or impacted on the study analysis are presented in an itemized fashion below (numbering and date of approval correspond to the one validated by the French regulatory authorities).

5.1 Initial version of the protocol

The initial version of the protocol aimed to include n=916 men with *de novo* mCSPC treated with ADT alone as SOC. With 458 allocated to ADT + abiraterone (arms B and D) and 458 allocated to ADT without abiraterone (arms A and C) and a total number of deaths of 406, a 4% level two-sided test of equal exponential survival would have an 80% power to detect a survival difference between abiraterone-treated and abiraterone-deprived arms with a constant hazard ratio of 0.75. With the same number of patients, and total number of 627 CRPC events, a 1% level two-sided test of equal exponential CRPC PFS would have an 84% power to detect an improvement in PFS with a hazard ratio of 0.75. The same hypotheses were made regarding the OS and PFS improvement related to the addition of prostate radiotherapy to systemic treatment.

5.2 Study amendments

Amendment N°14 (substantial; ethics committee approval: 07-OCT-2015)

Several studies¹⁻³ published after the initial approval of PEACE-1 protocol, established the benefit of supplementing androgen deprivation therapy (ADT) with docetaxel for diverse population of patients including *de novo* mCSPC patients, the ones enrolled in PEACE-1. Therefore, the study protocol was amended to authorize the use of docetaxel as part of the SOC, in patients fit to receive this treatment. Such a treatment came in effect in October 2015 at a time when 273 patients had already been included. For patients unfit to receive docetaxel, the SOC remained ADT alone. The stratification factor “docetaxel” (yes vs. no) was added to the randomization process. To adjust for the longer median overall survival mediated by the addition of docetaxel, the median time expected in the control arm was increased by 1 year, to 5 years, and the estimated follow-up duration after last patient entry until data maturity by 1 year bringing the expected study duration to 3 years of recruitment and 6.5 years of follow-up. The target hazard ratio and number of events (deaths) remained unchanged. The IDMC pointed out that the initial sample size planned (n=916) would be underpowered with the addition of a new group of patients (the docetaxel-treated) and that possible statistical interactions between the trial different components may reduce the power of the study on the overall population. The submission of a revised statistical plan/sample size was advised.

Amendment N°20 (substantial; ethics committee approval: 10-AUG-2017)

The sample size was increased to 1168 patients (+250 patients) based on simulations incorporating a statistical quantitative interaction between abiraterone and docetaxel. The clinical hypothesis was that the efficacy of abiraterone could be weaker in patients treated with ADT+docetaxel compared to ADT alone as SOC. Based on the LATITUDE⁴ and STAMPEDE⁵ trial results (presented at the American Society of Clinical Oncology (ASCO) meeting in June 2017) demonstrating that the upfront combination of abiraterone with ADT increased the overall survival of men with *de novo* metastatic prostate cancer (PEACE-1 study indication) compared to ADT alone, it was no longer possible for ethical reasons to allow these patients to be allocated to ADT alone when randomized to the SOC arm. Therefore, the protocol was amended to selectively restrict accrual to the patients which could be treated with ADT+docetaxel (866 patients were already included at this time).

Amendment N°23 (substantial; ethics committee approval: 22-MAR-2018)

In December 2017, two docetaxel-related deaths (one in each arm) occurred the same week at the same center. These events put the trial on hold for about three months, the time for this center to be audited by both the sponsor and health authorities. As both deaths were due to neutropenic fever it was decided as per the health authority requirement that the trial would be restarted on the condition that an amendment was made to enforce the mandatory prescription of G-CSF along with an

additional blood test around day 6 (for early neutropenia detection) for all docetaxel-treated patients until the trial completion. Once this amendment (N°23) was authorized on March 22, 2018 (210 patients were randomized afterwards), no subsequent toxicity-related death was observed.

Amendment N°32 (substantial; ethics committee approval: 13-JAN-2021)

Due to the Covid-19 pandemic, the submission and approval this amendment was considerably delayed. From April 2019 to April 2020 changes to be made to the analysis plan were discussed with the IDMC and the Steering Committee as some questions to be answered by PEACE 1 had been partially or totally solved. Specifically:

- The efficacy of abiraterone in hormone-sensitive metastatic prostate cancer treated by ADT alone as SOC was investigated by the LATITUDE study with regard to high-risk patients⁴, and the STAMPEDE study⁵
- The efficacy of prostate radiotherapy in hormone-sensitive metastatic prostate cancer treated by ADT alone as SOC was investigated by the HORRAD study⁶ and the STAMPEDE RT study⁷.

Therefore, once authorized, this amendment allowed us to:

- **Investigate a revise co-primary objective** to answer two questions of utmost interest for the medical community by:
 - Assessing the efficacy of abiraterone acetate and prednisone in patients with metastatic hormone-naïve prostate cancer treated by ADT and docetaxel (with or without prostate radiotherapy) in terms of overall survival (OS) and radiographic progression free survival (rPFS).
 - Assessing the efficacy of prostate radiotherapy in hormone-naïve prostate cancer patients with low metastatic burden (as defined in the CHARTED trial⁸), regardless of the SOC received (ADT with or without docetaxel, with or without abiraterone) in terms of OS and rPFS.
- **Focus on the timing of the final analyses instead of conducting intermediate analysis** based on the hypotheses that abiraterone in the ADT+ docetaxel population would have an efficacy and that prostate radiotherapy on the population with a low metastatic burden (in accordance with the revised primary objectives) would also have an efficacy.
- **Update the nature of the co-primary endpoint from PFS to radiographic PFS (rPFS):** In initial protocol PFS was based on rPFS and PSA progression events. However, post PEACE-1 initiation, various studies showed that PSA progression had poor prognosis value as a surrogate endpoint. In contrast, rPFS was demonstrated to be a valid surrogate endpoint for OS in metastatic prostate cancer and was an FDA-validated registration endpoint in hormone sensitive metastatic prostate cancer settings (PEACE-1 study indication). Hence, rPFS was requalified as a co-primary endpoint while PFS events already well defined and collected during the trial were used along with PSA progression to characterize and evaluate CRPC-free survival as a secondary endpoint.
- **Change the distribution of the type I error between the co-primary endpoints** in order to prioritize the overall survival analysis. Thus, the type I error for OS was set at 4.9% instead of 4%, and at 0.1% instead of 1% for rPFS.
- **Define the testing strategy** and decision rules for the fixed-sequence test in the study subpopulations to control the type I error and avoid multiplicity issues. Specifically, for the abiraterone question, the choice of testing first the overall population and then the ADT+docetaxel population was based on the following arguments: the trial had been designed for the population of men with de novo mCSPC with a SOC comprising ADT+/- docetaxel depending on the inclusion period. We felt that discarding the information provided by 40% of our study population (those receiving ADT alone as SOC) was unethical. To the contrary, keeping those patients in the analysis gave more power to the study (as more events were available in the overall population). The risk of being stopped after testing the overall population and not being able to present data for the ADT+docetaxel population was considered small, given the external efficacy data of abiraterone reported by the LATITUDE and STAMPEDE studies.

6. Supplementary data

6.1 Supplementary figures

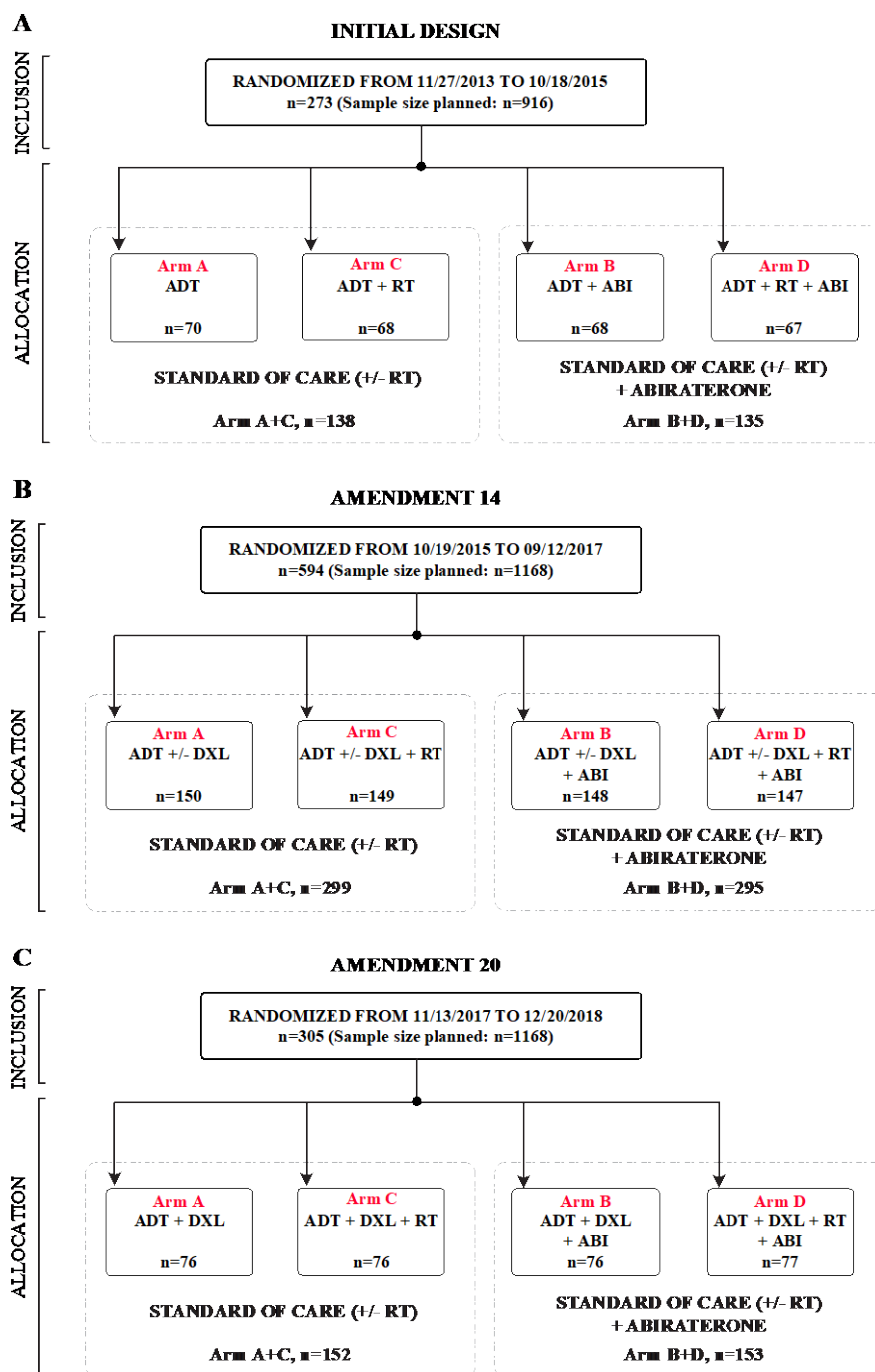


Figure S1. Randomization scheme per arm in response to metastatic castration-sensitive prostate cancer standard of care evolution. As per protocol, 916 men were initially to be equally randomized to arm A, B, C, and D (A). On October 7, 2015, the protocol amendment 14 authorized the use of docetaxel (B). As of August 10, 2017 amendment 20 enforced the

use of docetaxel for all included patients and increased the sample size to 1168 men (C). ABI stands for abiraterone, ADT for androgen-deprivation therapy, DXL for docetaxel, and RT for radiotherapy.

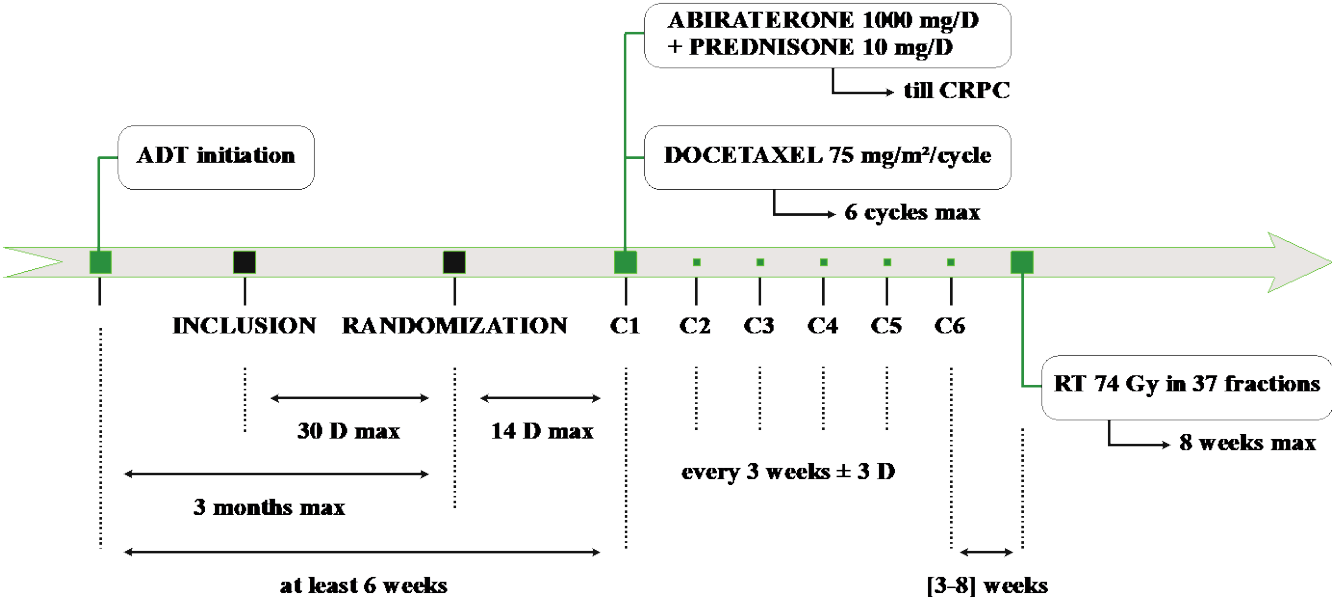


Figure S2. Treatment flow chart. ADT stands for androgen-deprivation therapy (induced by either GnRH agonist or antagonist or bilateral orchiectomy), C for cycle, CRPC for castration resistant prostate cancer, D for days, Gy for Gray, RT for radiotherapy. Treatments are depicted in green. Prednisone could be replaced by prednisolone. Cycles apply for the docetaxel regimen.

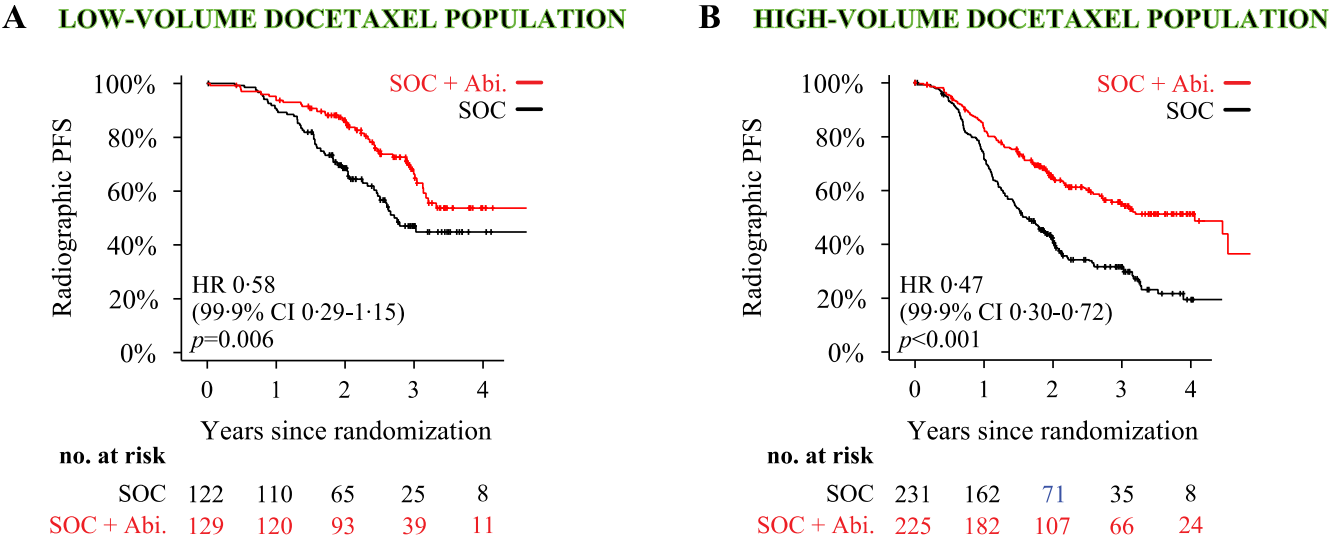


Figure S3. Kaplan-Meier estimates of the radiographic PFS within the ADT+docetaxel population. Time to event curves are presented for men with low (A) or high-volume (B) disease burden. SOC is ADT+Docetaxel+/-RT.

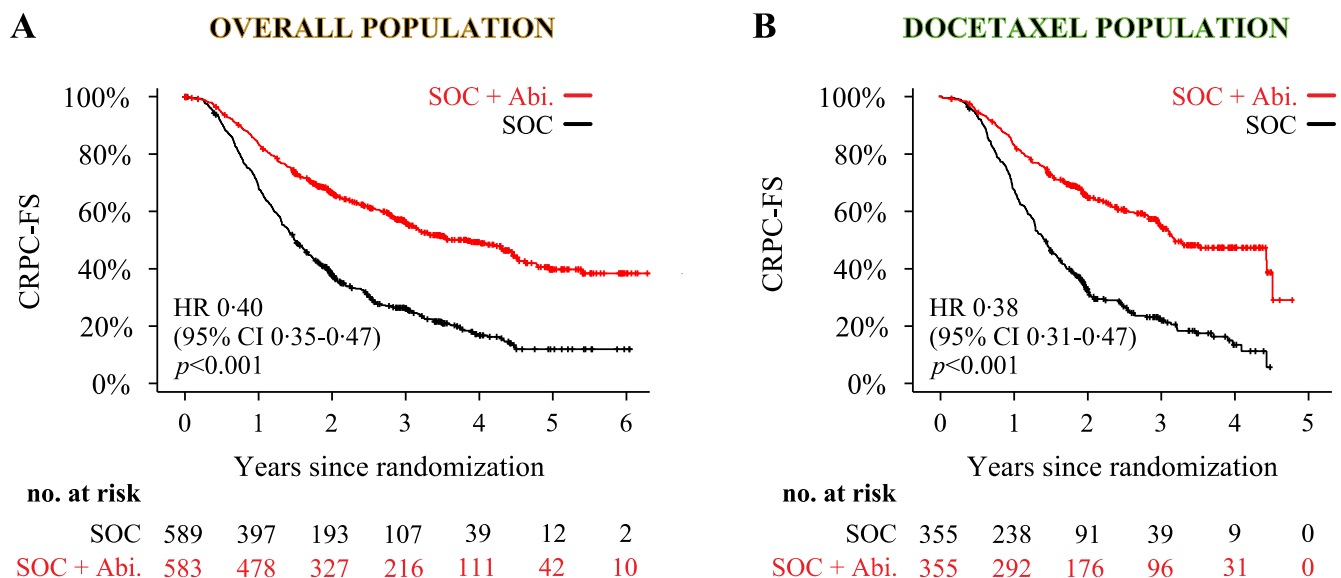


Figure S4. Kaplan-Meier estimates of Castration-Resistant Prostate Cancer Free Survival (CRPC-FS). Time to event curves are presented for the overall (A) and docetaxel populations (B).

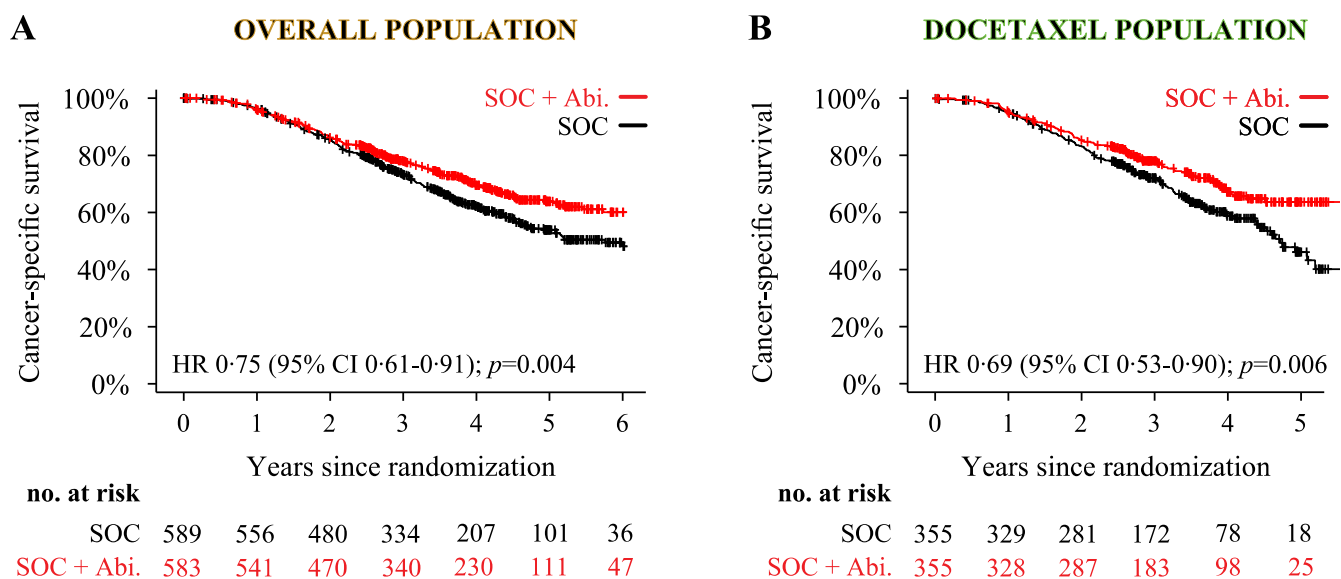


Figure S5. Kaplan-Meier estimates of Cancer-Specific Survival. Time to event curves are presented for the overall (A) and docetaxel populations (B).

6.2 Supplementary tables

Supplementary table 1: Treatment flow chart

	Baseline	Treatment Period									Follow up	
Examinations	Randomization	D1 ABI	C1D1 DTX*	C1D4 DTX	C1D7 DTX	CnD1	M6	M12	M18	M24 to CRPC	End of Treatment	Long Term Follow Up
Information Notice & Consent Form	X											
Inclusion/Exclusion criteria	X											
Randomization in TENALEA	X											
Medical history	X											
Disease diagnosis	X											
Physical examination												
Complete physical & vital signs	X	X				X	X	X	X	X	X	X
ECOG performance status	X	X				X	X	X	X	X	X	X
Concomitant treatments	X	X				X	X	X	X	X	X	X
Toxicities/AE collection		X				X	X	X	X	X	X ¹	X ¹
Laboratory tests												
Biochemistry	X ²	X ³⁺⁹	X ³			X ³⁺⁹	X ⁴⁺⁹	X ⁶⁺⁹	X ⁶⁺⁹	X ⁶⁺⁹	X ⁶	X ⁶
Hematology	X ²	X ³	X ³	X ¹⁵	X ¹⁵	X ³	X ⁴	X ⁶	X ⁶	X ⁶	X ⁶	X ⁶
Coagulation Factors & Serum Lipids	X ²	X ³				X ³	X ⁴	X ⁶	X ⁶	X ⁶	X ⁶	X ⁶
Serum testosterone & PSA	X ²	X ³				X ³	X ⁴⁺⁵	X ⁶	X ⁶	X ⁶	X ⁶⁺¹²	X ⁶
Paraclinical assessments												
Electrocardiogram	X ¹³										X	
Radiological assessment	X ⁸⁺¹³	X ⁷⁺⁸				X ⁷⁺⁸	X ⁷⁺⁸	X ⁷⁺⁸	X ⁷⁺⁸	X ⁷⁺⁸	X ⁸	
Bone densitometry	X						X	X		X ¹⁴		
Study drug												
Abiraterone Acetate + Prednisone		X				X	X	X	X	X		
Quality of life questionnaires												
FACT-P	X						X	X	X	X	X	
BPI-SF	X	X ¹⁰				X ¹⁰	X	X	X	X	X	
QLQ-C30	X						X		X		X	
Ancillary study												
Tumor biopsy collection	X											
Blood collection	X	X ¹¹				X ¹¹				X ¹¹		

ABI stands for Abiraterone; BPI-SF for Brief Pain Inventory Short Form; C for cycle; D for Days; DTX for Docetaxel; FACT-P for Functional Assessment of Cancer Therapy-Prostate; M for Month; QLQ-C30 for the EORTC core Quality of Life Questionnaire-C30.

* If different than D1 abiraterone

1 : Serious Adverse Events (SAEs) to be collected until 30 days after the last administration of abiraterone or docetaxel (or 90 days after the last radiotherapy treatment). Delayed SAEs related to study drug and/or research to be reported until end of study.

2 : Laboratory assessments to be done at the latest 2 days before randomization.

3 : Laboratory assessments to be done at C1D1, C2D1, C3D1, C4D1, C5D1, C6D1 (maximum 2 days before).

4 : Window allowed +/- 2 days.

5 : PSA to be dosed at M8.

6 : Window allowed +/- 1 month.

7 : Radiological assessment to be done only if disease progression is biologically or clinically diagnosed or suspected.

8 : Bone scan (any technique) **and** CT scan or MRI, same type of examinations to be used from baseline to end of study of patient.

9 : For patients randomized in arms B and D, AST, ALT and bilirubin should be measured every two weeks for the first three months (hence at C2D14 and C3D14), at M5, and then every month from M6 to end of treatment visit.

10 : Quality of life questionnaire to be filled out at M2.

11 : To be collected at M1, M3 and at progression.

12 : PSA only.

13 : To be done within 3 months of randomization.

14 : M24 only.

15 : Hematology assessments to be done **at C1D4 and C1D7 (+/-1 day): hemoglobin, platelet count, neutrophil count, red blood cell count, white blood cell count**

Supplementary Table 2. Treatments given post-progression in the ADT+docetaxel population – ITT population

At least one treatment, n (%)	SOC + Abi CRPC n=141	SOC CRPC n=263
Life-prolonging treatment	104 (74)	221 (84)
Second-generation hormonal therapy	65 (46)	213 (81)
Abiraterone	22 (16)	153 (58)
Enzalutamide	57 (40)	119 (45)
Docetaxel	29 (21)	25 (10)
Cabazitaxel	84 (60)	114 (43)
Radium 223	3 (2)	11 (4)
Lu-PSMA	2 (1)	3 (1)

CRPC stands for Castration-Resistant Prostate Cancer; Lu-PSMA, Lutetium-177–Prostate-Specific Membrane Antigen; SOC, Standard of Care

Supplementary Table 3. Reason for treatment discontinuation – Safety population

	Treatment allocation			
	ADT+DXL population (SOC = ADT + DXL +/- RT)		ADT population (SOC = ADT +/- RT)	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D	Arm A+C	Arm B+D	Arm A+C
	(n=347)	(n=350)	(n=226)	(n=237)
Safety population	183 (164)	252 (98)	138 (88)	163 (74)
Available (non-available)				
Reason				
Second cancer	1 (<1)	3 (1)	5 (4)	4 (3)
Disease progression	124 (67)	241 (95)	88 (64)	148 (89)
Patient's decision	11 (7)	4 (2)	4 (3)	5 (3)
Death	13 (7)	2 (<1)	9 (7)	4 (3)
Toxicity/adverse event	32 (17)	1 (<1)	29 (21)	1 (<1)
Other	2 (1)	1 (<1)	3 (2)	1 (<1)

Supplementary Table 4. Nature of fatal (Grade 5) adverse events in the safety population

	Treatment allocation			
	ADT+DXL population (SOC = ADT + DXL +/- RT)		ADT population (SOC = ADT +/- RT)	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D	Arm A+C	Arm B+D	Arm A+C
	(n=347)	(n=350)	(n=226)	(n=237)
Safety population	7	3	8	5
Reason				
Second cancer	2		3	2
Cardiovascular disease	1		2	2
Sepsis	2	1		
Pulmonary disease	1	2		
Other	1		3	1

Supplementary Table 5. Severe adverse event rate per 100 subject-years

	Treatment allocation			
	ADT+DXL population (SOC = ADT + DXL +/- RT)		ADT population (SOC = ADT +/- RT)	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D	Arm A+C	Arm B+D	Arm A+C
	(n=347)	(n=350)	(n=226)	(n=237)
Adverse events				
Number of severe (Grade 3-5) AE	454	341	290	157
Number of patients with at least 1 severe (Grade 3-5) AE, n (%)	217 (63)	181 (52)	149 (66)	97 (41)
Number of subject-years	926	622	789	528
Severe adverse event rate per 100 subject-years	49	55	37	30

Supplementary table 6. Expected frequent (>5%) severe (Grade 3-5) adverse events – Safety population

	Treatment allocation			
	ADT+DXL population		ADT population	
	(SOC = ADT + DXL +/- RT)		(SOC = ADT +/- RT)	
	SOC + Abi	SOC	SOC + Abi	SOC
	Arm B+D	Arm A+C	Arm B+D	Arm A+C
Safety population	(n=347)	(n=350)	(n=226)	(n=237)
Expected frequent adverse events				
Severe (Grade 3-5) adverse events				
Hypertension	76 (22)	45 (13)	66 (29)	38 (16)
Neutropenia	34 (10)	32 (9)	0 (0)	0 (0)
Hepatotoxicity	20 (6)	2 (1)	14 (6)	3 (1)
Febrile neutropenia	18 (5)	19 (5)	2 (1)	1 (0)
Gamma-glutamyltransferase increased	17 (5)	14 (4)	6 (3)	4 (2)
Erectile dysfunction	7 (2)	5 (1)	12 (5)	13 (5)
Blood alkaline phosphatase increased	15 (4)	12 (3)	6 (3)	13 (5)
Expected fatigue events				
(Grade 3-5) events	10 (3)	15 (4)	3 (1)	0 (0)
(Grade 1-2) events	234 (67)	242 (69)	83 (37)	61 (26)
Expected peripheral neuropathy events				
(Grade 3-5) events	4 (1)	6 (2)	1 (0)	0 (0)
(Grade 1-2) events	136 (39)	119 (34)	7 (3)	4 (2)

7. The question of radiotherapy efficacy

The 2x2 factorial design of PEACE-1 offers the possibility to answer two independent questions based on different hypothesis: the question of Abiraterone efficacy and the question of Radiotherapy efficacy. However, this manuscript solely reports on the abiraterone question as the efficacy and safety data related to the radiotherapy question will only be analyzed once the pre-planned numbers of rPFS and OS events are reached in the low-volume disease population of mCSPC men.

Indeed, the 2013 initial protocol planned to analyze the efficacy of abiraterone and radiotherapy simultaneously as the effect of these treatments was assumed to be identical ($HR_{ABI}=HR_{RT}=0.75$).

However, late 2018 and early 2019, the HORRAD and STAMPEDE RT trials results were published. Those trials assessed the efficacy of prostate radiotherapy in the same setting than PEACE-1 (patients with de novo metastatic hormone-sensitive prostate cancer treated by ADT alone as SOC). Neither, the HORRAD trial (n=432) nor the STAMPEDE RT trial (n=2061) found evidence of an overall survival (OS) benefit in the overall population (HR: 0.90, 95% CI 0.70–1.14, and HR: 0.92, 95% CI 0.80–1.06; $p=0.266$, respectively) but raised the possibility that survival might be improved in a subgroup of patients with low metastatic burden (HR: 0.68, 95% CI 0.42–1.10 and HR: 0.68, 95% CI 0.52–0.90; $p=0.007$, respectively).

In light of these results, we chose to amend the protocol (Amendement n°32 cf Appendix - § 5. Amendments to the study design and statistical analysis plan) and revise our primary objective to confirm HORRAD and STAMPEDE RT findings regarding the efficacy of prostate radiotherapy in patients with low metastatic burden. The timing of the analysis is based on the number of events in this subgroup (299 radiographic progressions or deaths for the rPFS analysis and 213 deaths for the OS analysis). So far, these predefined numbers of events have not been reached in the PEACE1 patients with low metastatic burden. We had anticipated this situation in the protocol, which in the chapter “9.3. Interim and final analysis” p51 reads: “It is likely that the analyses of OS for the abiraterone question and the prostate radiotherapy questions will be performed separately due to different onset of deaths in the subgroups of interest (ADT+docetaxel population for abiraterone, that of low metastatic burden population for prostate radiotherapy)” and in the chapter “9.5.2 Analysis of Efficacy endpoints” p52 reads: “If the analysis of prostate radiotherapy and that of abiraterone are not performed at the same time, analysis of abiraterone will be adjusted on radiotherapy but the $HR_{RADIO THERAPY}$ will not be made public”.

Consequently, the radiotherapy efficacy question will be analyzed and published in due course.

8. Supplementary references

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